

Portfolio 3: Functions

BMI Calculator

Description

This program demonstrates the use of functions in Python by creating a simple BMI Calculator. The user will input their name, weight, and height. The program will compute the BMI and determine the BMI category.

Concepts Used

- Functions
- Conditional Statements
- Variables
- User Input
- Arithmetic Operations

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In [13]: # Portfolio 3: Functions
# BMI Calculator

# Function to compute BMI
def compute_bmi(weight, height):
    bmi = weight / (height ** 2)
    return bmi

# Function to determine BMI category
def bmi_category(bmi):
    if bmi < 18.5:
        return "Underweight"
    elif bmi < 25:
        return "Normal Weight"
    elif bmi < 30:
        return "Overweight"
    else:
        return "Obese"

# Function to give health advice
def health_advice(category):
    if category == "Underweight":
        return "Eat more healthy food."
    elif category == "Normal Weight":
        return "Keep maintaining your health!"
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elif category == "Overweight":
    return "Exercise regularly."
else:
    return "Consult a healthcare professional."

# Function to display results
def display_result(name, age, gender, bmi, category, advice):

# Function Calls
    bmi = compute_bmi(weight, height)
    category = bmi_category(bmi)
    advice = health_advice(category)

# Display Output
display_result(name, age, gender, bmi, category, advice)

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In [9]: # User Inputs
name = input("Enter your name: ")
age = int(input("Enter your age: "))
gender = input("Enter your gender (Male/Female): ")

weight = float(input("Enter your weight in kilograms: "))
height = float(input("Enter your height in meters: "))

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In [15]: #Output

print(" BMI RESULT ")
print("-----")
print("Name:", name)
print("Age:", age)
print("Gender:", gender)
print("BMI:", round(bmi, 2))
print("Category:", category)
print("Health Advice:", advice)

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BMI RESULT
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Name: Tresha Mae Madrid
Age: 18
Gender: Female
BMI: 20.28
Category: Normal Weight
Health Advice: Keep maintaining your health!

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